Parallel Batch Breadth-First Search: Framework and Applications

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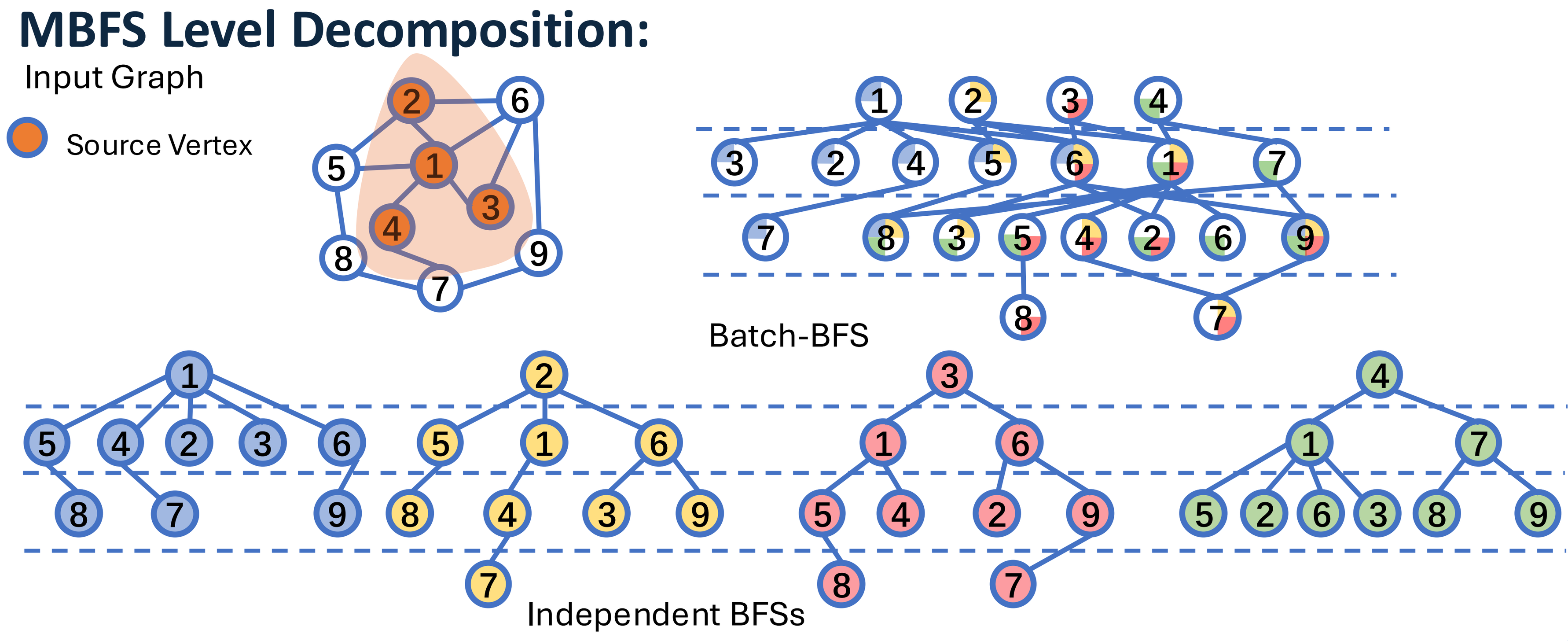
Definition of Multi-BFS

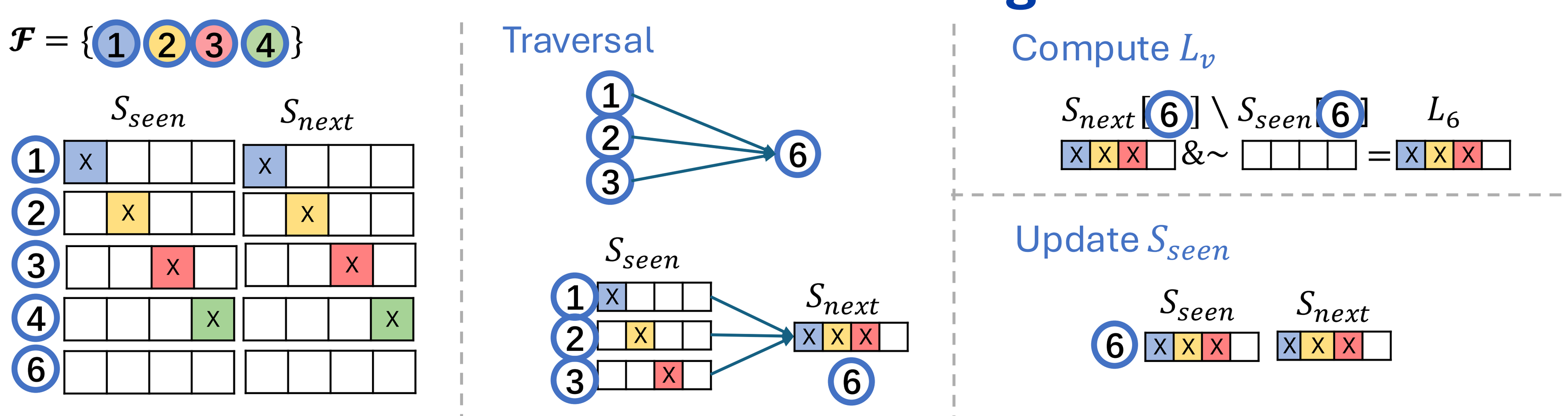
- **Multi-BFS (MBFS):** conducts BFSs from multiple sources
- **MBFS Level Decomposition:**



Related Work

Algorithms	Applications	Parallelism	However
Cluster BFS [2]	Cluster landmark-labeling	intra	Only supports clusters
Ligra-Ecc [3]	k-BFS Eccentricity	intra	Only computing Ecc
MS-BFS[4]	Closeness Centrality	inter	Lack parallelism when sources are small
MITra [5]	BFS + Dijkstra + SpMV + …	inter	
Ours	MBFS + CBFS + Eccentricity + Closeness + …	intra	

Our Parallel Batch-BFS Algorithm



Experiment Setup

- **Systems**: **96 cores** with hyper-threads, **1.5 TB memory**
- **Graphs**: **18 undirected graphs**, which are either **social or web graphs** with low diameters

Batch-BFS Experiments

- **Microbenchmark:**
  - Seq: the standard textbook single-source BFS
  - InterPar: each thread runs a sequential single source BFS
  - Ligra[1]: parallel single-source BFS
  - MSBFS[4]: multi-source BFS (sequential on each thread)

64 Random Sources

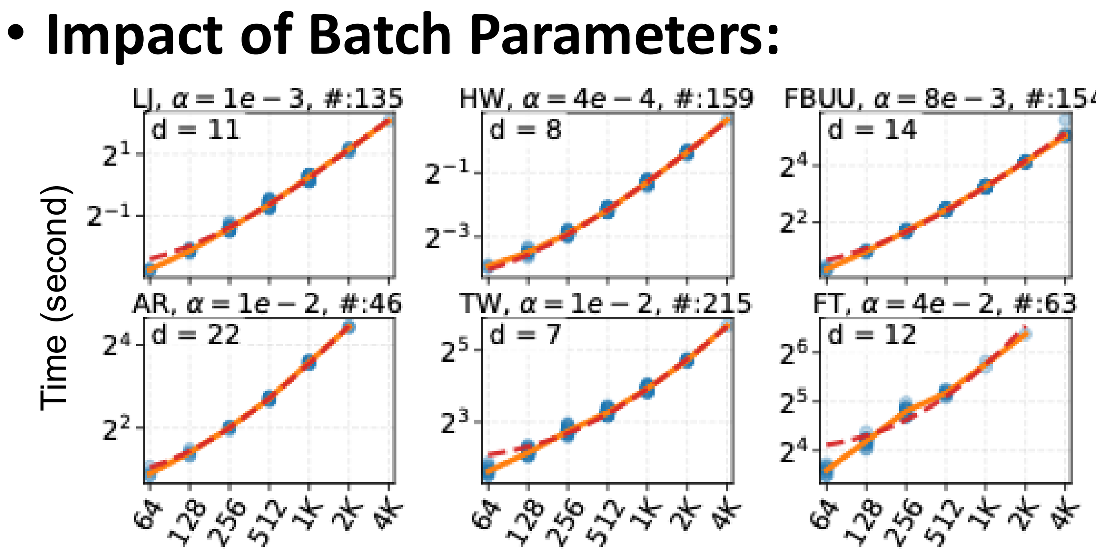
Alg.	Running Time (s) Seq	MSBFS	Ligra	Inter	Ours	Speedup Over Seq MSBFS	Ligra	Inter	Ours
Bit	✗	✓	✗	✗	✓	✓	✗	✗	✓
Inter	✗	✗	✗	✓	✗	✗	✗	✓	✗
Intra	✗	✗*	✓	✗	✓	✗*	✓	✗	✓
EP	0.15	0.02	0.05	.005	**.004**	7.75	2.85	28.7	**37.5**
SLDT	0.17	0.02	0.05	.006	**.004**	8.22	3.44	30.9	**44.1**
DBLP	0.66	0.08	0.12	0.02	**0.01**	8.02	5.47	31.5	**62.5**
YT	2.99	0.33	0.19	0.09	**0.02**	9.05	15.6	32.2	**123**
SK	5.94	0.64	0.23	0.21	**0.04**	9.33	25.4	28.9	**160**
IN04	3.16	0.93	0.93	0.09	**0.08**	3.39	3.41	34.1	**37.4**
LJ	36.8	3.75	0.73	1.42	**0.12**	9.82	50.7	25.9	**317**
HW	16.9	0.96	0.23	0.48	**0.05**	17.5	72.7	34.9	**332**
FBUU	226	23.8	5.64	7.84	**0.95**	9.50	40.1	28.8	**237**
FBKN	149	23.5	4.59	5.12	**0.91**	6.33	32.4	29.1	**163**
OK	62.4	4.84	0.68	2.11	**0.13**	12.9	92.0	29.5	**470**
INDO	29.5	8.22	3.05	0.85	**0.51**	3.60	9.68	34.9	**57.9**
EU	96.6	6.78	4.41	3.37	**0.36**	14.2	21.9	28.7	**270**
UK	74.3	12.8	3.24	2.14	**0.68**	5.82	22.9	34.7	**109**
AR	112	22.9	14.6	3.11	**1.50**	4.91	7.71	36.2	**75.1**
TW	867	78.3	6.27	41.6	**2.53**	11.1	138	20.9	**343**
FT	2227	327	12.1	105	**8.99**	6.81	184	21.2	**248**
SD	1730	233	21.0	68.8	**6.79**	7.41	82.4	25.1	**255**
MEAN	27.8	3.50	1.23	0.94	**0.20**	7.95	22.7	29.5	**140**

64 Highest-Degree Sources

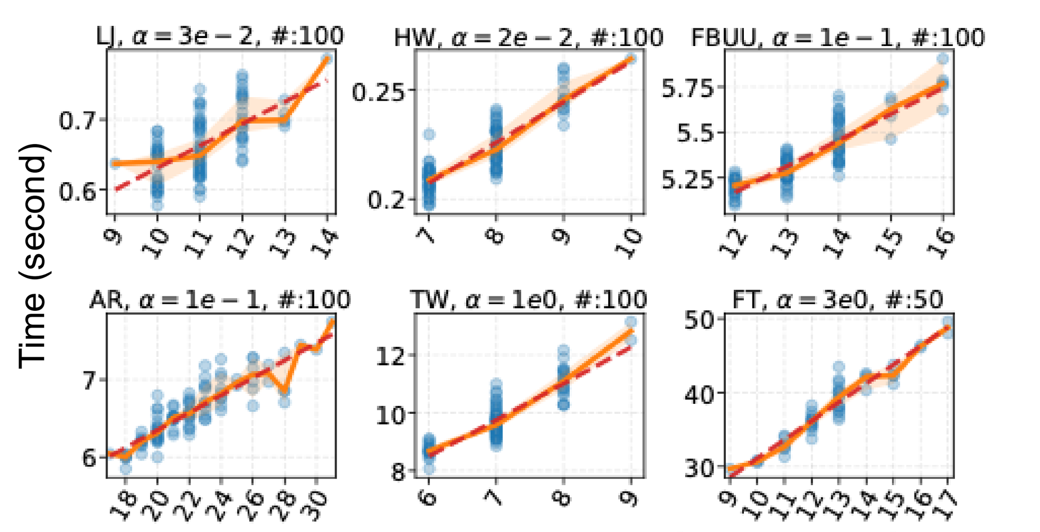
Alg.	Running Time (s) Seq	MSBFS	Ligra	Inter	Ours	Speedup Over Seq MSBFS	Ligra	Inter	Ours
Bit	✗	✓	✗	✗	✓	✓	✗	✗	✓
Inter	✗	✗	✗	✓	✗	✗	✗	✓	✗
Intra	✗	✗*	✓	✗	✓	✗*	✓	✗	✓
EP	0.15	0.01	0.05	.005	**.004**	10.3	3.21	29.2	**33.5**
SLDT	0.17	0.01	0.06	.006	**.006**	12.8	3.06	27.5	**30.5**
DBLP	0.65	0.07	0.12	0.02	**.008**	9.78	5.55	30.9	**83.0**
YT	3.00	0.20	0.18	0.09	**0.01**	15.3	16.4	32.0	**213**
SK	5.93	0.44	0.24	0.19	**0.02**	13.5	24.5	31.0	**248**
IN04	3.23	0.53	0.89	0.09	**0.05**	6.07	3.62	34.3	**71.1**
LJ	39.3	2.51	0.86	1.38	**0.07**	15.7	45.7	28.4	**526**
HW	17.0	0.54	0.23	0.48	**0.03**	31.2	72.8	35.6	**675**
FBUU	243	19.9	5.48	7.96	**0.75**	12.2	44.4	30.6	**325**
FBKN	159	16.7	4.61	5.10	**0.68**	9.51	34.5	31.2	**235**
OK	61.8	3.60	0.73	2.06	**0.09**	17.2	84.9	30.1	**713**
INDO	30.5	3.72	3.03	0.81	**0.21**	8.20	10.1	37.6	**143**
EU	102	7.06	4.54	3.79	**0.36**	14.5	22.5	26.9	**282**
UK	77.0	8.60	3.19	2.16	**0.36**	8.95	24.1	35.7	**213**
AR	112	13.8	14.5	2.99	**0.81**	8.14	7.72	37.5	**138**
TW	909	35.6	6.47	42.2	**1.25**	25.5	140	21.5	**728**
FT	2356	127	13.6	104	**4.11**	18.5	173	22.6	**573**
SD	1843	173	26.2	67.6	**5.50**	10.6	70.4	27.3	**335**
MEAN	28.5	2.26	1.26	0.94	**0.13**	12.6	22.6	30.2	**216**

Seq / 1x — 50x — >100x

- **Impact of Batch Parameters:**



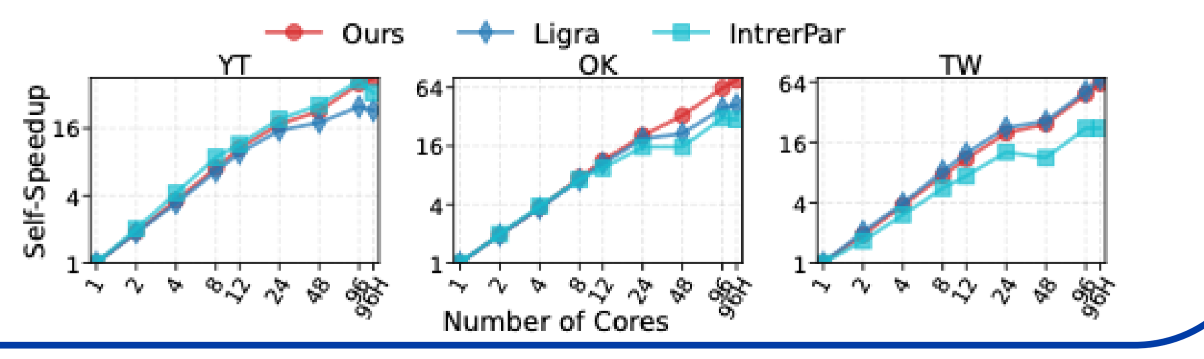
Running time vs. batch diameter d with $k = 512$ sources.



Running time vs. batch size k with fixed d

- **Scalability**

The self-speedup on different number of cores. Higher is better.

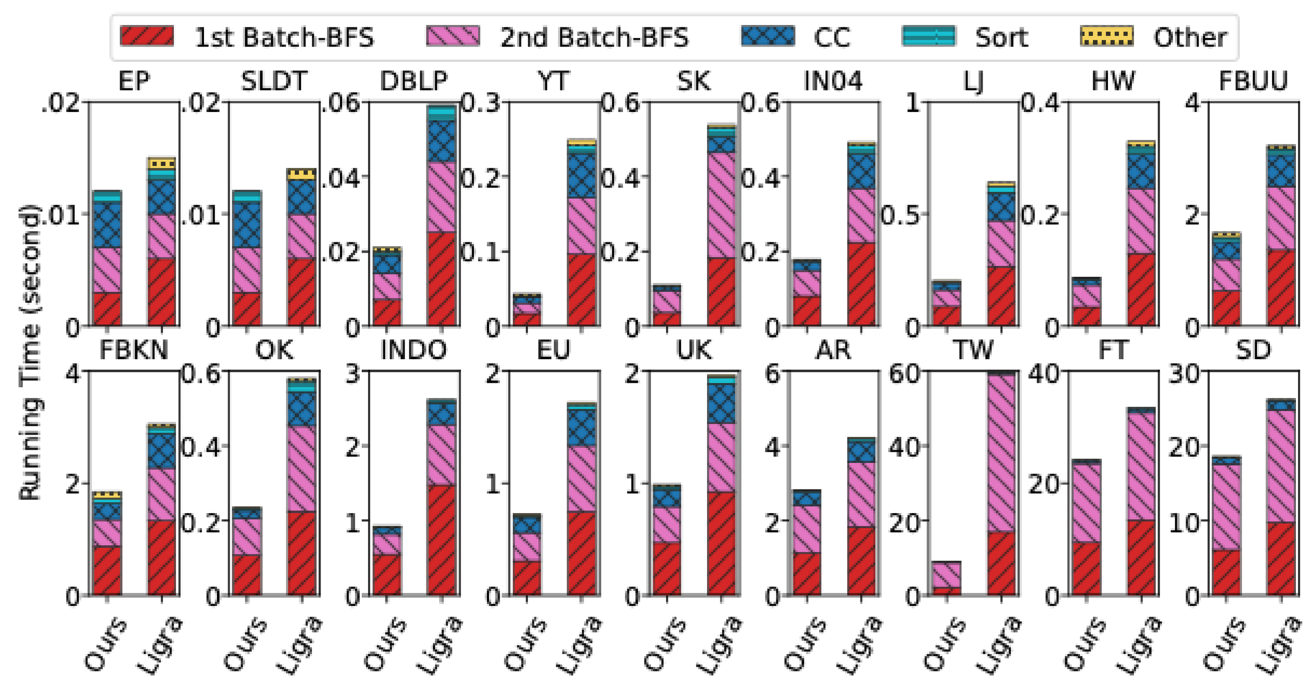


Applications

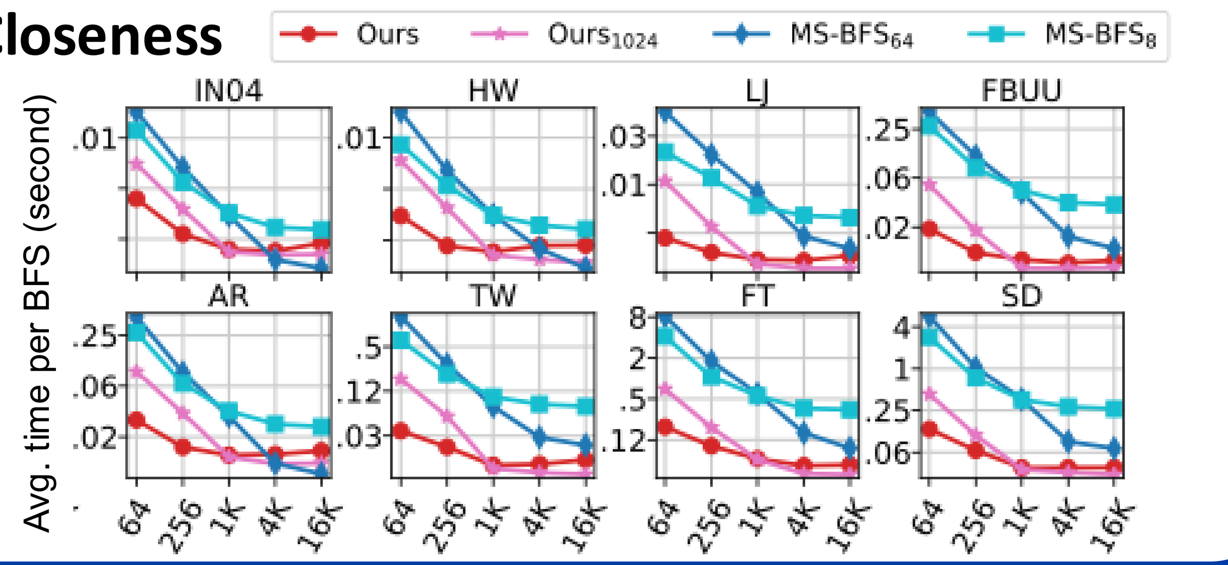
- **Landmark Labeling**

Data	Normal LL Index Time (s) Ligra	Ours	T_{ligra}/T_{ours}	ϵ (%)	Cluster LL Index Time (s) CBFS	Ours	T_{CBFS}/T_{ours}	ϵ (%)
EP	1.00	**0.01**	100	0.40	**0.02**	0.03	0.67	0.08
SLDT	0.94	**0.01**	94.0	0.72	**0.02**	**0.02**	1.00	0.15
DBLP	3.31	**0.05**	66.2	2.53	**0.08**	0.10	0.80	2.22
YT	8.31	**0.13**	63.9	0.31	**0.21**	0.26	0.81	0.30
SK	11.7	**0.25**	46.8	1.38	0.57	**0.55**	1.04	0.68
IN04	20.2	**0.40**	50.5	2.13	1.07	**0.94**	1.14	1.90
LJ	35.9	**0.75**	47.9	4.98	**1.75**	1.77	0.99	4.30
HW	8.53	**0.23**	37.1	10.6	1.00	**0.67**	1.49	5.63
FBUU	145	**7.80**	18.6	6.18	**11.6**	14.8	0.78	11.9
FBKN	129	**7.31**	17.6	6.21	**10.9**	13.6	0.80	11.9
OK	25.9	**0.87**	29.8	8.68	3.01	**2.87**	1.05	7.71
INDO	78.6	**2.49**	31.6	3.16	6.31	**5.06**	1.25	1.53
EU	96.7	**2.86**	33.8	2.65	7.86	**7.18**	1.09	1.28
UK	78.2	**3.79**	20.6	3.90	9.48	**8.10**	1.17	4.90
AR	273	**6.63**	41.2	2.61	20.4	**17.4**	1.17	4.04
TW	112	**9.51**	11.8	1.49	31.6	**25.6**	1.23	1.36
FT	257	**31.8**	8.08	16.7	62.8	**54.6**	1.15	12.3
SD	368	**33.5**	11.0	0.59	77.6	**70.7**	1.10	0.36
MEAN	32.4	**1.00**	32.4	2.50	2.33	**2.28**	1.02	1.85

- **Eccentricity**



- **Closeness**



References

[1] Julian Shun and Guy E. Blelloch. 2013. Ligra: A Lightweight Graph Processing Framework for Shared Memory. 135–146.

[2] Letong Wang, Guy Blelloch, Yan Gu, and Yihan Sun. 2025. Parallel Cluster-BFS and Applications to Shortest Paths. In Algorithm Engineering and Experiments (ALENEX). SIAM, 42–55.

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[4] Manuel Then, Moritz Kaufmann, Fernando Chirigati, Tuan-Anh Hoang-Vu, Kien Pham, Alfons Kemper, Thomas Neumann, and Huy T Vo. 2014. The more the merrier: Efficient multi-source graph traversal. Proceedings of the VLDB Endowment (PVLDB) 8, 4 (2014), 449–460.

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